

Data Visualization

ECTS

5

Prerequisites

Students taking the course are expected to have basic programming skills (e.g. IT-PRO1 or IT-PRG1 and IT-WEB1).

Main purpose

This course provides students with an introduction to the fundamentals of data visualization.

Students will learn how to transform data into graphical representations, explore data relationships, and visually communicate insights using D3.js.

Knowledge

After completing this course, it is expected that the student will be able to:

- Understand theoretical concepts of data visualization, including:
 - models for design
 - systematic approaches to implementation.
- Describe different data types and their implications for the chosen visualization.
- Describe various visualization types
- Reflect on key principles of data-related visual communication.

Skills

After completing this course, it is expected that the student will be able to:

- Use valid data preprocessing techniques.
- Apply appropriate visualization methods based on the data type and requirements.
- Implement custom visualizations using D3.js.

Competences

After completing this course, it is expected that the student will be able to:

- Utilize visualization to analyse datasets.
- Develop visualization strategies for presenting data insights.
- Evaluate visualization projects in terms of design, communicative impact, and implementation.

Teaching methods and study activities

This course is organized into 12 sessions.

Each course begins with a theoretical introduction to different aspects of data visualization. This will lead to a hands-on workshop related to the introduced theory.

A number of hand-ins based on the lesson content will be part of the final exam.

Exam

Exam prerequisites:

None.

Exam type:

20-minute oral examination based on one of the hand-ins that have been developed throughout the semester.

Hand-ins must be submitted by the deadline.

At the exam, the student draws one of the known hand-ins. The examination starts from the student's solution and then broadens into an assessment of the rest of the course syllabus.

Assessment is solely based on the student demonstrating the expected knowledge at the oral examination.

Internal assessment.

Tools allowed:

All tools and resources introduced throughout the course.

Re-exam:

Same as the ordinary exam.

Grading criteria

Grading based on the Danish 7-point scale.

Responsible

Rachel Stoklund (rsto)

Valid from

01-08-2026

Course type

Elective for Software Engineering.

Elective for the specialisation Machine Learning and AI Engineering as well as the specialization Business Information Systems

Valgfag for Softwareingeniør.

Valgfag for Machine Learning and AI Engineering specialisering samt for Business Information Systems specialisering